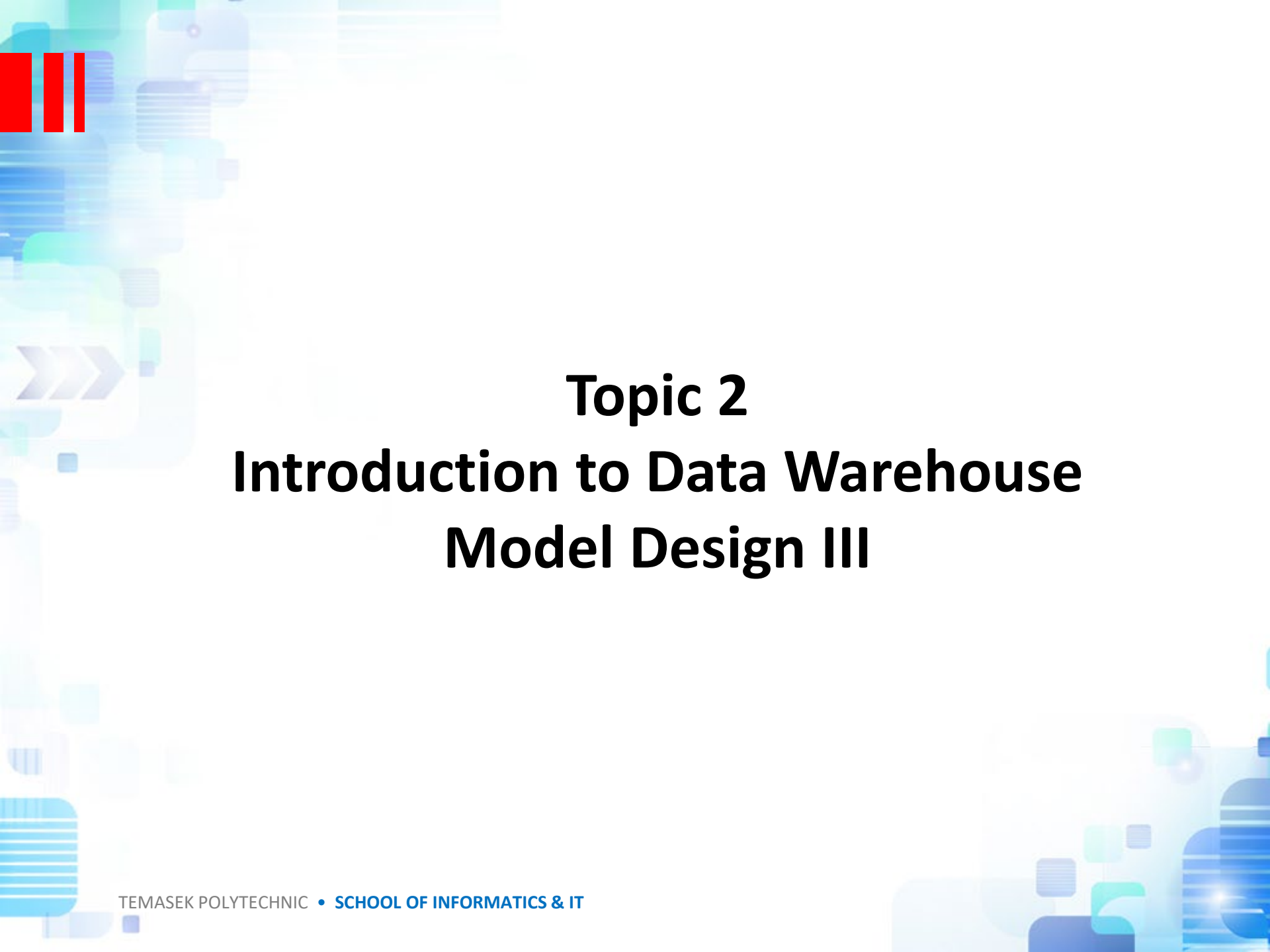




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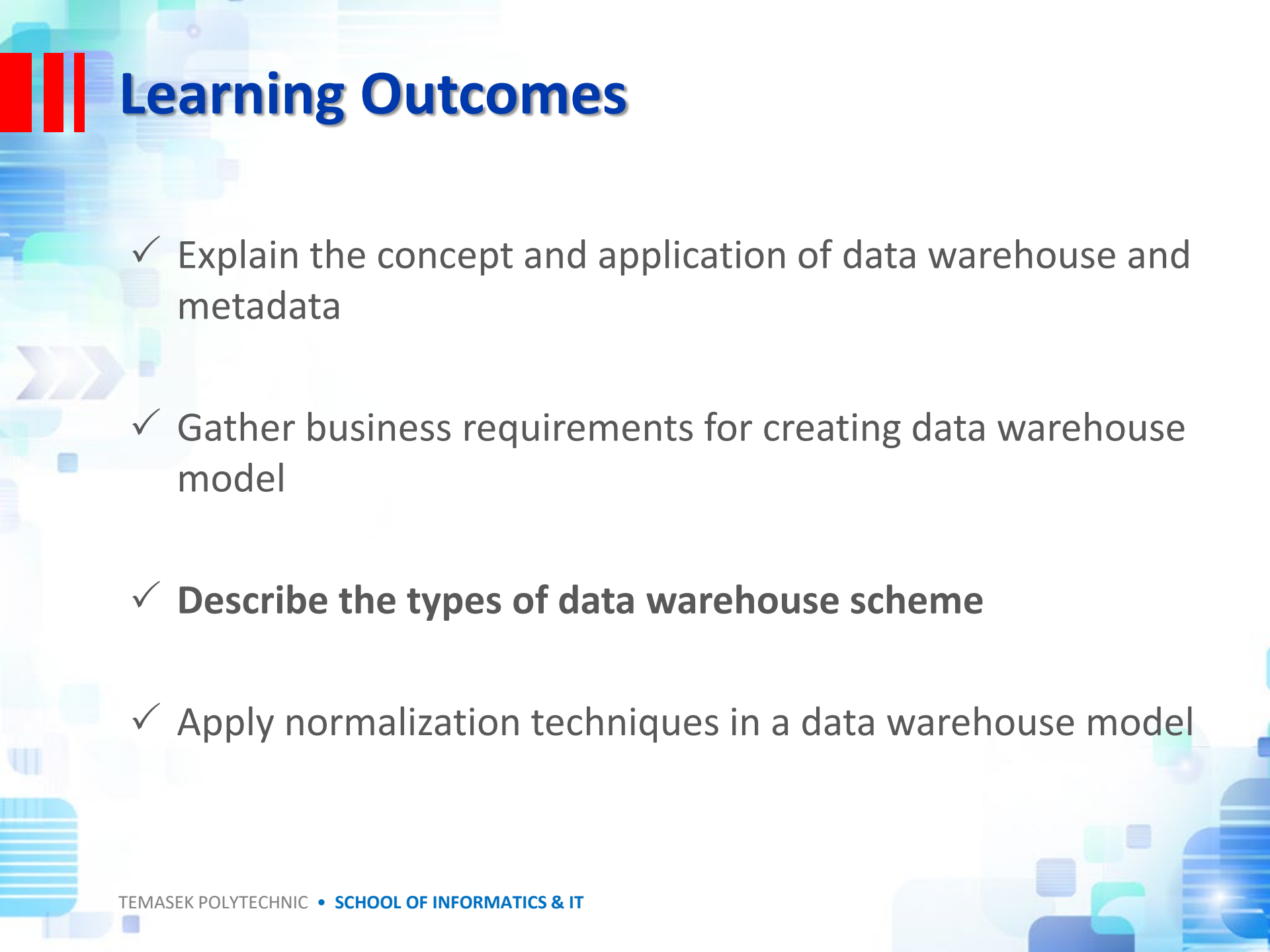
Data Warehousing and Business Intelligence

CDA2C01



Topic 2

Introduction to Data Warehouse Model Design III



Learning Outcomes

- ✓ Explain the concept and application of data warehouse and metadata
- ✓ Gather business requirements for creating data warehouse model
- ✓ **Describe the types of data warehouse scheme**
- ✓ Apply normalization techniques in a data warehouse model



DATA WAREHOUSE SCHEMAS

Dimensional Nature of Business Data

Managers think of the business in terms of business dimensions. The following example shows the kind of questions managers are likely to ask for decision making:

Marketing Vice President: “Give me the revenue by Product”, “Give me the revenue by Geographic region”, “Give me the revenue by Customer demographic (age, income level, occupation...)”

The marketing VP is not satisfied with just a single number for revenue. He wants the revenue numbers broken down by month, division, customer demographic, sales office, product category etc. These are the business dimensions along which he wants to analyse the numbers.

Dimensional modeling comprises of dimensions, facts and measures.

Elements of Dimensional Modeling

only 1 fact table but can have multiple dimension table

Fact

- It is a collection of associated data items, consisting of measures and context data. It typically represents business items or business transactions.

Dimensions

- It is a collection of data which describe one business dimension. Dimensions decide the contextual background for the facts, and they are the framework over which OLAP is performed.

Measure

- It is a numeric attribute of a fact, representing the performance or behavior of the business relative to the dimensions.

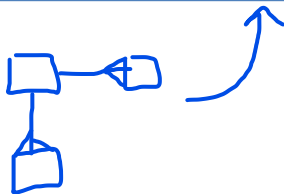
ER Modelling versus Dimensional Modelling

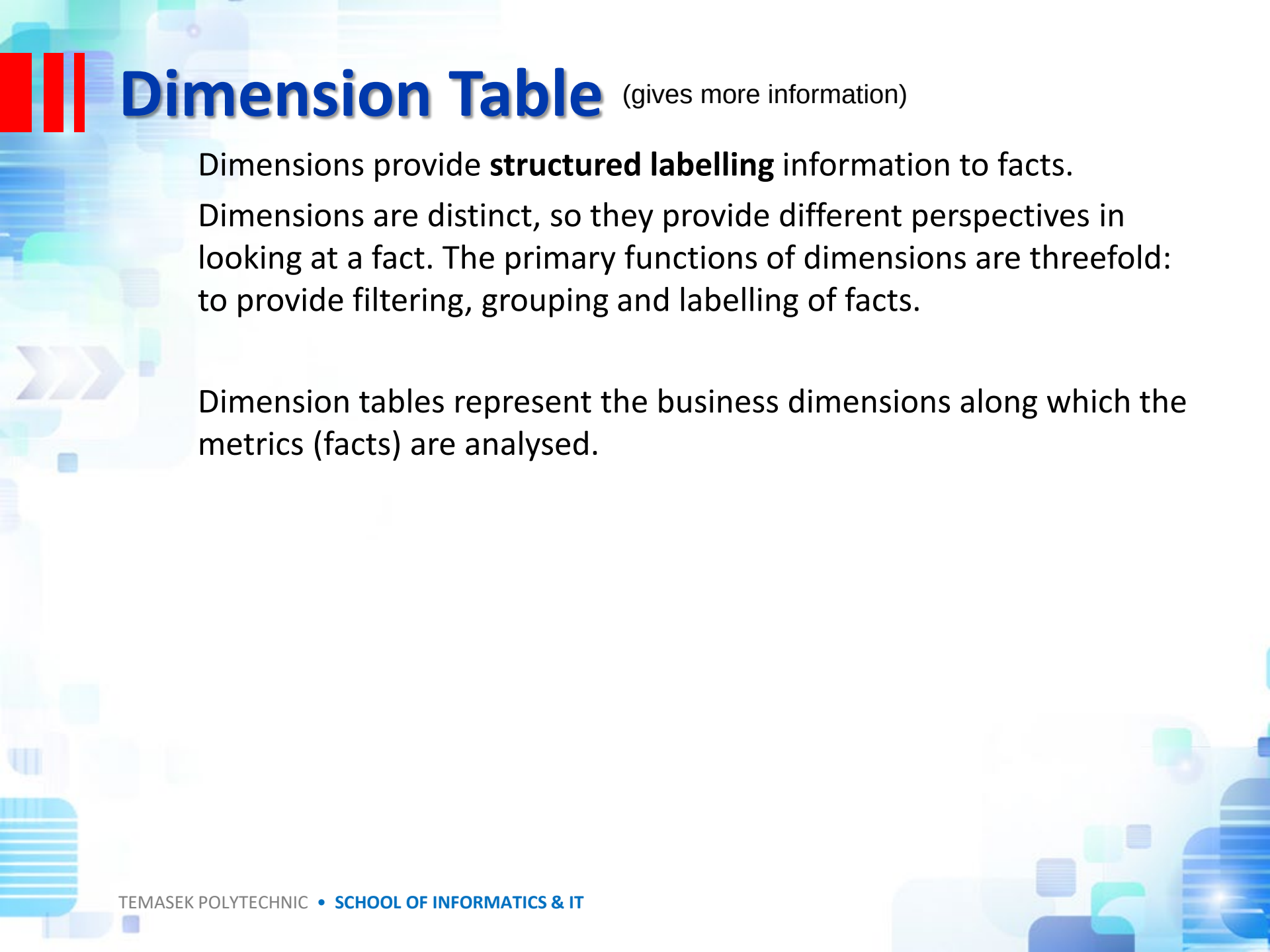
Entity-Relationship Modelling

- OLTP systems capture details of events or transactions
- OLTP systems focus on individual events
- An OLTP system is a window into micro-level transaction
- Picture at detail level necessary to run the business
- Suitable only for questions at transaction level
- Data consistency, non-redundancy and efficient data storage critical

Dimensional Modelling

- DW meant to answer questions on overall process
- DW focus is on how managers view the business
- DW reveals business trends
- Information is centered around a business process
- Answers show how the business measures the process
- The measures to be studied in many ways along several business dimensions





Dimension Table (gives more information)

Dimensions provide **structured labelling** information to facts.

Dimensions are distinct, so they provide different perspectives in looking at a fact. The primary functions of dimensions are threefold: to provide filtering, grouping and labelling of facts.

Dimension tables represent the business dimensions along which the metrics (facts) are analysed.

Dimension Table Characteristics

Dimension Table Characteristics

Dimension Table Key

Table is wide

Textual Attributes

Attributes Not Directly Related

Fewer Records

Flattened out, Not Normalised

Drilling Down, Rolling Up

Multiple Hierarchies

CUSTOMER

customer_key

name

age

gender

customer_id

billing_address

billing_city

billing_state

billing_postal_code

shipping_address

Dimension
table

Dimension Table Characteristics

Dimension Table Key

The primary key of the dimension table uniquely identifies each row in the table.

Table is Wide

A typical dimension table has many columns or attributes (>50), hence we say that the table is wide.

Textual Attributes

Attributes of the dimension table are usually text. Numerical values are uncommon.

Attributes Not Directly Related

Some attributes are not directly related to other attributes of the same dimension table. E.g. package size is not directly related to product brand, but both attributes could be present in the PRODUCT dimension table.

Fewer Records

A dimension table typically has fewer rows than a fact table.

no new record everyday (eg. only add a new record when it is a new customer)

Dimension Table Characteristics

Not Normalised

One way to improve query performance is to reduce the number of JOINS between tables. To this end, the dimension table in the star schema is not further broken down into smaller tables (normalisation). We say that the dimension table is flattened out.

Drilling Down, Rolling Up

The attributes in a dimension table provide the ability to get to the details (drilldown), or generate summary information (rollup). For example, a sales fact can be drilled down by time: YEAR > QUARTER > MONTH, or location: COUNTRY > STATE > BRANCH.

Multiple Hierarchies

Drilling down and rolling up can occur along multiple hierarchies in a single dimension table. For example, a PRODUCT dimension table can drill down by CATEGORY > BRAND > SKU, or MANUFACTURER > FACTORY > PRODUCTION BATCH.

Fact Table

A **fact** is a value or measurement meaningful to the business. Also known as **key business metrics**.

Examples of facts: Sales, Margin dollars, Quantity sold, Number of defects in a production batch (manufacturing), Number of rooms occupied in a hotel, Number of customer complaints

Fully Additives measures are attributes that may be summed up by simple addition. E.g. quantity_order

Semi Additives measures are derived attributes where the values cannot be add up to come up with aggregated number. E.g. margin_percentage.

Fact Table Characteristics

Fact Table Characteristics

Concatenated key

Data Granularity

Fully Additive Measures

Semi-Additive Measures

Table Deep, not wide

Sparse Data

Degenerate Dimension

ORDER FACTS
product_key
order_date_key
salesperson_key
customer_key
order_dollars
extended_cost
margin_dollars
quantity_ordered
order_number*
order_line*

Fact Table

Fact Table Characteristics

Granularity, or the **data grain**, refers to the level of detail in the fact table.

The higher granularity, the finer the grain, the more detailed is the fact.

The lower granularity, the coarse the grain, the less detailed is the fact.

E.g.: Product category is low granularity, and coarse grain, while stock-keeping unit (SKU) is high granularity, and fine grain.

The granularity of data can also refer to the size in which data fields are sub-divided.

For example, a postal address can be recorded, with coarse granularity, as a single field:

- address = 200 2nd Ave. South #358, St. Petersburg, FL 33701-4313 USA

Or with fine granularity, as multiple fields:

- street address = 200 2nd Ave. South #358
- city = St. Petersburg
- state = FL
- postal code = 33701-4313
- country = USA

Fact Table Characteristics

Fully Additive Measures

Measures in the fact table that can be aggregated to provide a meaningful answer.

Examples of measures in aggregation queries:

- Total revenue (SUM)
- Average order amount
- Highest quantity ordered (MAX)

Semi-Additive Measures

Measures in the fact table that may not always make sense in an aggregation query.

E.g.

Average sales margin percentage makes sense but summing up the sales margin percentage for all products do not.

Fact Table Characteristics

Table Deep, Not Wide

A typical fact table contains fewer attributes than a dimension table, but a much larger number of records.

Sparse Data

A considerable number of rows in the fact table may not have any value (NULL) for some measures. This is referred to as sparse data.

E.g.

No orders are taken on public holidays. Hence on these dates, the order_amount attribute in the fact table is NULL.

Fact Table Characteristics

Degenerate Dimensions

The attributes order_number and order_line are not measures or facts. After selecting attributes for fact and dimension tables, you may be left over with data elements that are neither fact nor dimension attributes.

E.g. reference numbers like order numbers, invoice numbers, order line numbers etc.

Such attributes are called degenerate dimensions and they are kept as attributes of the fact table.

```
ORDER_FACTS
product_key
order_date_key
salesperson_key
customer_key
order_dollars
extended_cost
margin_dollars
quantity_ordered
order_number*
order_line*
```

A Typical Fact Table

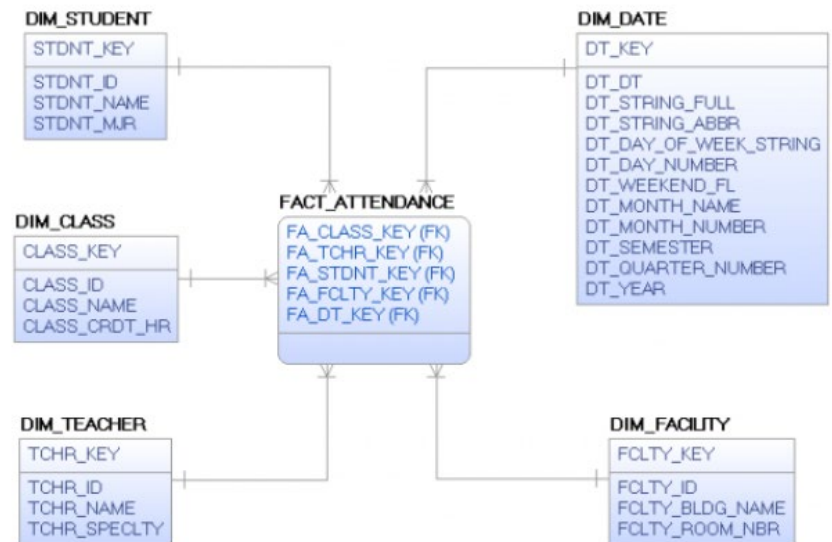
*Degenerate Dimensions

Fact Table Characteristics

Factless Fact Table

Situation arises when the fact table represent events. Row presence indicate the presence of the event. Such fact tables does not need to contain facts. They are factless fact table.

Tracks the attendance of student.



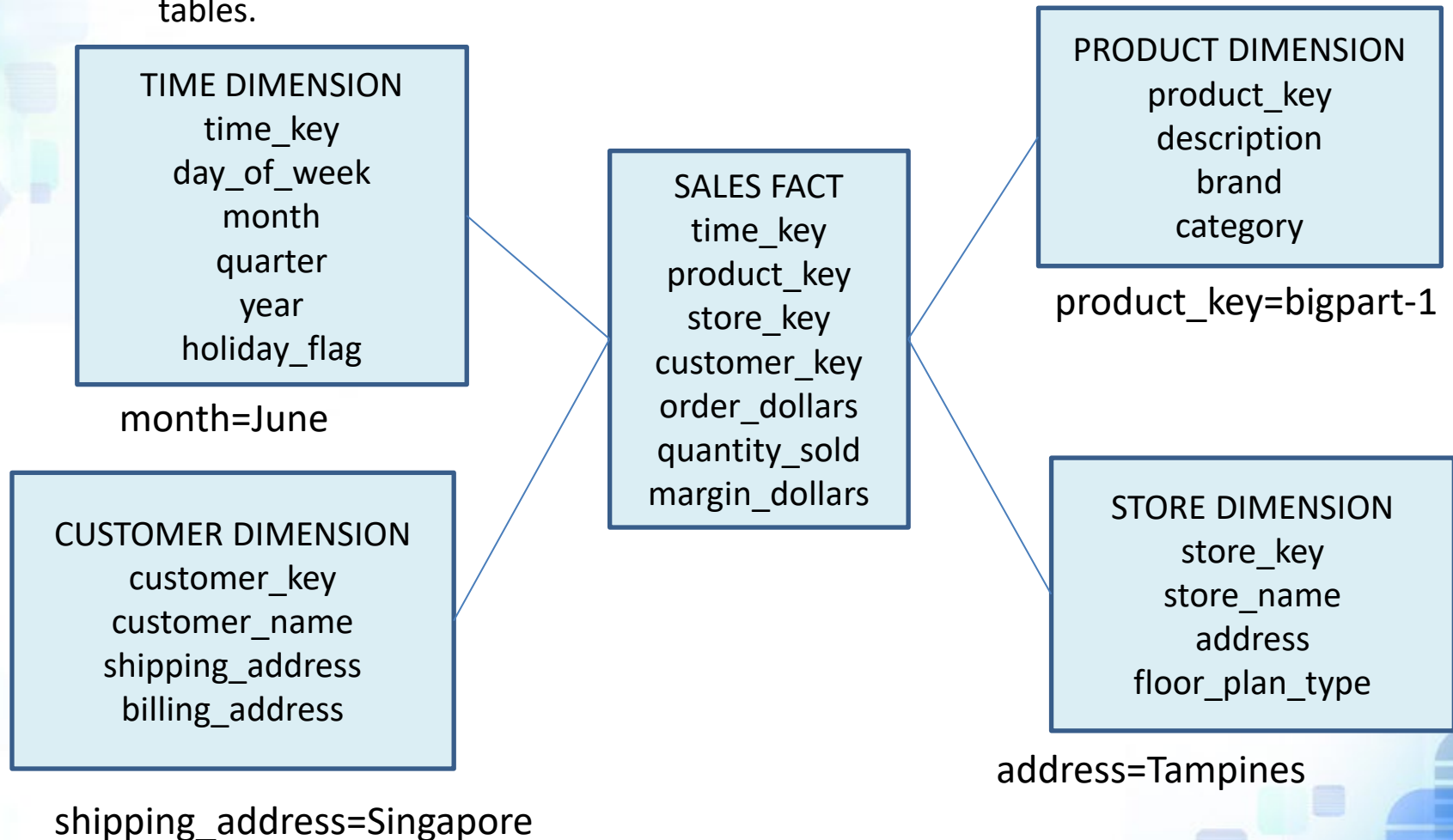
The Star Schema

- The dimensional model can take the form of a star schema.
- There is one large dominant table in the centre of the schema. It is the only table in the schema with multiple joins connecting it to other tables. The other tables all have only a single join attaching them to the central table.
- Central table is known as the **fact table**.
- The other tables are known as the **dimension tables**.
- The star schema can be easily understood by users because the structure mirrors how they normally view their metrics along business dimensions.

Advantages	Disadvantages
Easy Understand by Users	Not fully normalised
Better Query Performance	

A Typical Star Schema

The schema is called a star schema because the diagram resembles a star. The fact table is at the centre of the star, and the pointed edges of the star are the dimension tables.



Querying The Star Schema

- When a query is made against the star schema, the results of the query are produced by joining one or more dimension tables with the fact table i.e. a row in the fact table is joined with another row from the dimension table.
- The next slide shows an example of a simple query – What is the quantity sold and order dollars for product *bigpart-1*, relating to customers in *Singapore*, obtained by store branches in Tampines, in the month of *June*?

Advantages of the Star Schema

Easy for Users to Understand

- The star schema reflects how the users think and need data for querying
 - Metrics are the facts (fact table)
 - Analyse the metrics along business dimensions (dimension tables)
- The star schema is easy to navigate. Navigation is limited to the relationships between the fact and dimension tables.

Better Query Performance

- The star schema is not fully normalised, so there are less tables.
- Less tables means less JOINS in a query, thereby increasing query performance (faster reads)

Disadvantages of the Star Schema

The star schema is not fully normalised, so there is a higher risk of data redundancy. Can you think of an example of redundancy in the dimension tables? What values are likely to be repeated?

Drilldown/Rollup Queries

Another type of queries is the drilling down of summary numbers to get at the details. Aggregation (“roll-up”) queries are the opposite of drilldown queries.

Are the following queries drilldown or rollup?

- Find the revenue (order dollars) broken down by specific product or financial quarter.
- Find the total revenue for the “big parts” brand in the year 2013.

Product dimension

Brand=big parts

Product key=bigparts-1

Product key=bigparts-2

...

Time dimension

Year=2013

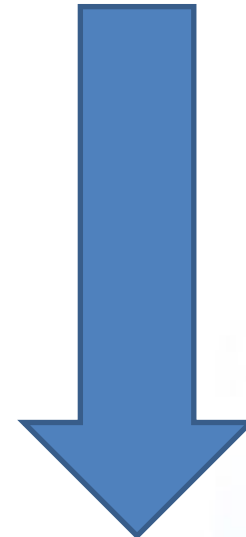
Quarter=1

Quarter=2

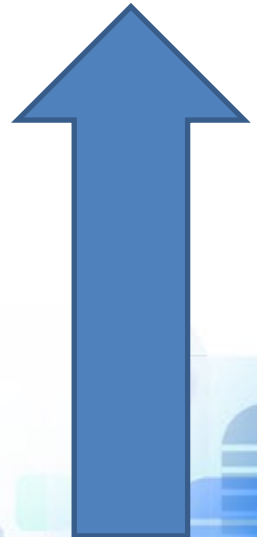
Quarter=3

Quarter=4

Drilldown



Rollup



Surrogate Keys

Do not use operational system keys as primary keys for dimension tables. Otherwise, a dependency between the operational system and the data warehouse will be created. If the operational system key has to be changed one day, the dimension table will also be affected by this change.

The answer is to use **surrogate keys**, which are simply system-generated unique identifiers. These keys are generated in the data warehouse. Unlike the operational system keys, surrogate keys do not have any built-in meanings.

The surrogate keys will be mapped to the operational system keys. The general practice is to keep the operational system keys as additional attributes in the dimension tables.

Surrogate Keys

Example – Product dimension

The stock-keeping unit (SKU) id is the operational system key that assigns an unique identifier to a particular product.

The data warehouse should not use SKU as the primary key of the Product dimension table. Instead, the data warehouse can generate its own surrogate key, which is the `product_key`. The surrogate key is mapped to the SKU as each row in the Product dimension table has both attributes.

PRODUCT DIMENSION

`product_key`

SKU

description

brand

category

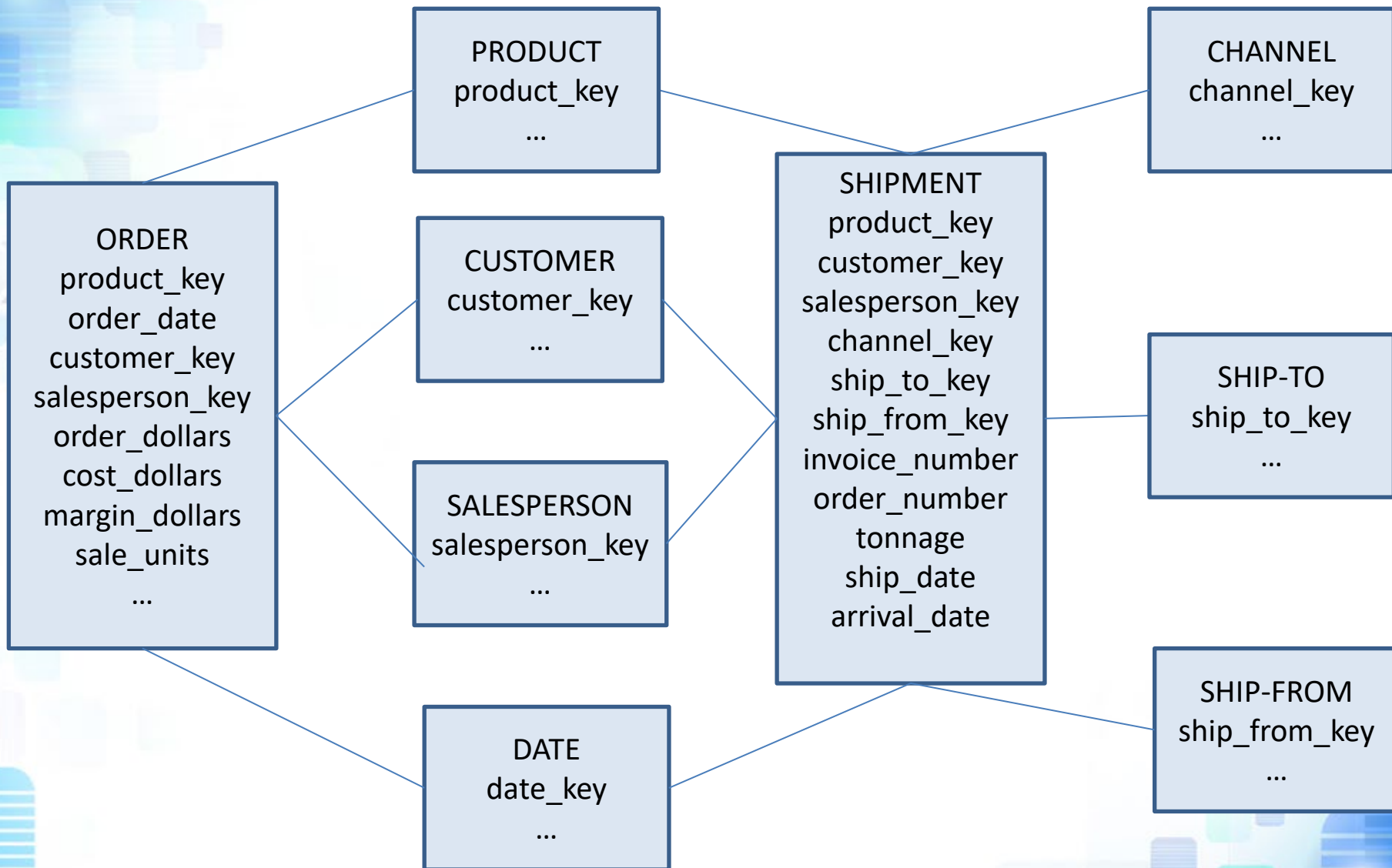
Conformed Dimensions

Conformed dimensions are dimension tables that are **shared** by different fact tables. Conformed dimensions form common links between star schemas.

Conformed dimensions enforce consistency. For example, if the product dimension is shared between two fact tables of sales and inventory, then the attributes of the product dimension must have the same meaning in relation to each fact table.

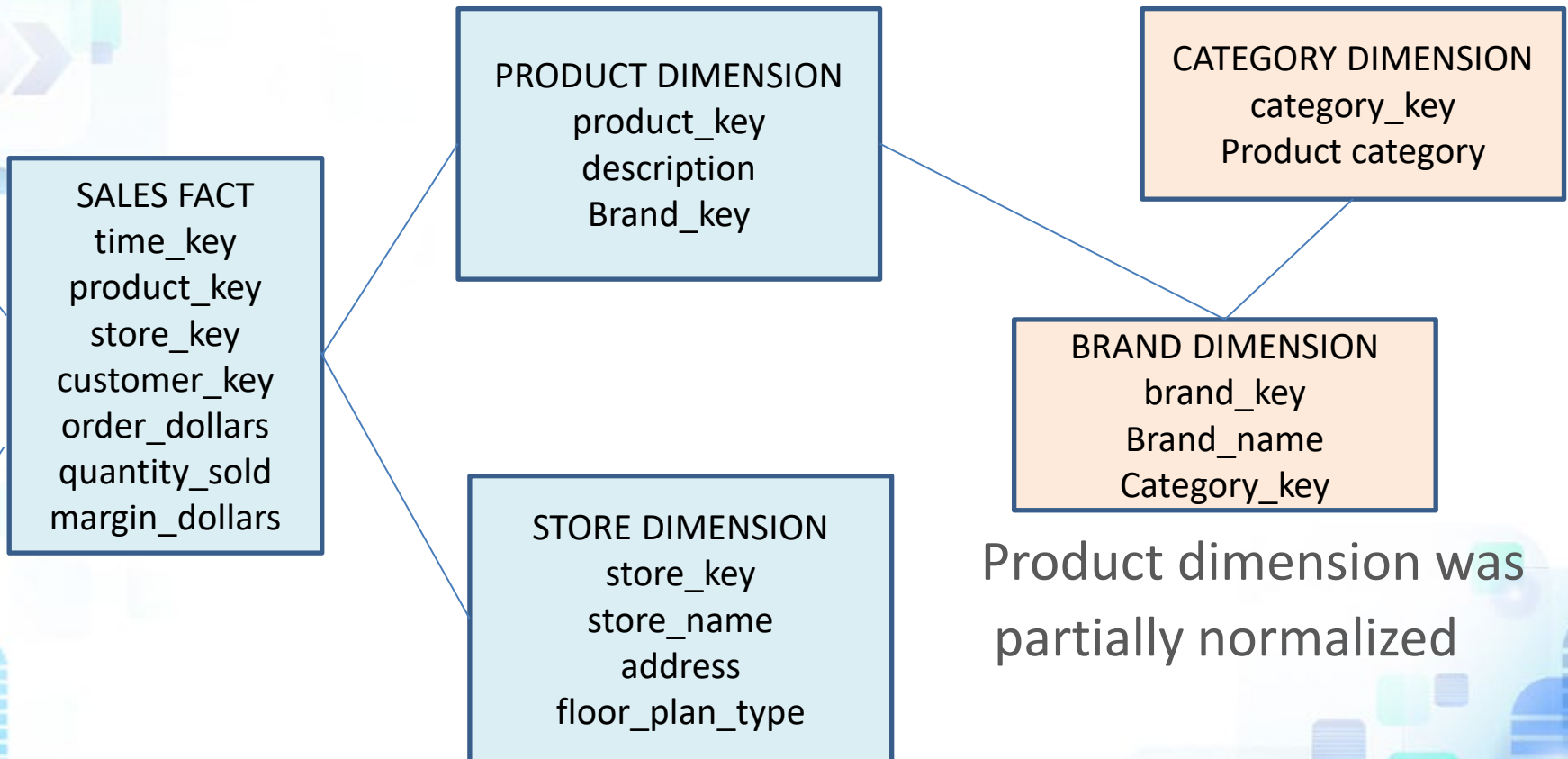
Conformed dimensions allow querying of multiple data marts. Conformed dimensions form the 'bus' of the data mart bus data warehouse architecture.

Conformed Dimensions (example)



Snowflake Schema

- "Snowflaking" is a method of normalizing the dimension tables in a STAR schema.



Product dimension was partially normalized

Advantages and Disadvantages of Snowflaking

Advantages

- Small savings in storage space
- Normalized structures are easier to update and maintain

Disadvantages

- Schema less intuitive and end-users are put off by the complexity
- Ability to browse through the contents difficult
- Degraded query performance because of additional joins

Snowflaking is **not generally recommended** in a data warehouse environment. Query performance takes the highest significance in a data warehouse and snowflaking hampers the performance.